

FEODOSIA, Ukraine

WMO: 339760

Lat: 45.042N

Lon: 35.382E

Elev: 26

StdP: 101.01

Time Zone: 3.00 (E03)

Period: 82-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.4	-8.8	-15.2	1.0	-10.5	-12.7	1.3	-8.3	12.3	-5.3	11.3	-2.0	4.7	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	31.1	21.8	29.8	21.5	28.6	21.0	23.3	28.6	22.5	27.9	21.8	27.3	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.3	26.8	20.6	15.3	26.1	19.7	14.5	25.2	69.6	28.5	66.4	28.0	63.7	27.4	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	8.6	7.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	3.3	0.7	5.1	10.7	16.8	20.7	23.8	23.9	18.7	13.2	6.9	3.5		
	DBStd	9.00	4.72	5.89	4.37	2.88	3.58	2.57	2.53	2.49	3.26	4.37	5.47	4.68		
	HDD10.0	997	208	263	158	23	2	0	0	0	1	19	118	204		
	HDD18.3	2684	466	495	411	231	72	7	1	1	34	165	343	459		
	CDD10.0	1858	1	2	6	43	213	322	428	431	263	120	26	3		
	CDD18.3	502	0	0	0	1	25	79	171	173	46	8	0	0		
	CDH23.3	3306	0	0	0	1	91	364	1294	1351	192	13	0	0		
	CDH26.7	833	0	0	0	0	8	65	353	387	20	1	0	0		
(14) Wind	WSAvg	3.0	3.5	3.4	3.2	2.9	2.7	2.6	2.7	2.6	2.7	3.0	3.2	3.3		
(15) Precipitation	PrecAvg	444	35	31	30	35	36	44	36	48	36	25	37	47		
	PrecMax	649	102	77	108	85	122	232	122	183	152	78	104	107		
	PrecMin	263	1	4	4	0	1	2	3	1	0	4	2	2		
	PrecStd	98	24	18	22	23	28	40	27	41	38	19	24	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.4	16.6	21.7	27.6	30.5	32.2	33.1	28.7	24.9	19.0	15.0		
		MCWB	9.1	9.7	11.2	14.1	18.9	22.0	23.0	22.2	21.0	18.9	15.9	12.7		
	2%	DB	10.9	11.0	13.7	19.0	25.4	28.4	30.9	31.5	27.0	22.7	16.8	12.4		
		MCWB	8.6	8.5	9.4	13.1	18.4	21.2	22.0	21.7	20.6	18.2	14.2	9.9		
	5%	DB	10.0	9.5	12.2	16.8	23.7	26.9	29.9	30.1	25.7	20.8	15.1	10.8		
		MCWB	7.9	7.6	8.8	12.1	17.3	20.3	21.6	21.4	19.4	17.3	12.6	8.9		
	10%	DB	9.1	8.0	10.7	15.3	22.3	25.3	28.7	28.7	24.2	19.1	13.7	9.5		
		MCWB	7.4	6.5	8.0	11.4	16.8	19.4	21.1	20.9	18.7	16.0	11.6	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.1	11.7	15.2	20.1	23.2	24.9	24.4	22.7	20.0	16.6	13.0		
		MCDB	11.2	12.9	15.6	20.5	25.8	28.5	30.7	30.6	27.1	23.9	18.5	14.7		
	2%	WB	8.7	8.8	10.1	13.7	19.1	22.1	23.7	23.0	21.3	18.8	14.5	10.3		
		MCDB	10.4	10.9	13.1	17.7	24.0	27.2	28.7	28.8	25.8	22.1	16.5	12.1		
	5%	WB	8.2	7.7	9.2	12.6	18.1	21.1	22.7	22.3	20.1	17.7	12.9	9.0		
		MCDB	9.8	9.3	11.6	16.0	22.6	25.6	27.9	28.3	24.7	20.3	14.4	10.6		
	10%	WB	7.6	6.6	8.3	11.7	17.3	20.2	21.9	21.6	19.1	16.3	11.9	8.1		
		MCDB	9.0	8.0	10.6	14.7	21.5	24.4	27.4	27.6	23.3	18.9	13.4	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.4	5.5	5.9	7.1	7.1	7.8	8.3	7.9	6.7	4.9	4.8		
		MCDBR	4.7	6.1	7.5	8.3	8.7	9.0	9.0	9.5	9.1	7.8	6.5	5.3		
	5% WB	MCWBR	3.5	4.7	4.9	4.8	4.7	4.7	4.5	4.6	5.0	4.7	4.5	4.1		
		MCWBR	4.7	5.8	6.9	7.4	8.0	7.6	8.1	8.4	8.3	7.1	5.8	5.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.348	0.377	0.406	0.440	0.412	0.415	0.436	0.438	0.411	0.383	0.359	0.347		
		taud	2.329	2.249	2.195	2.135	2.310	2.343	2.296	2.277	2.320	2.356	2.359	2.331		
	Ebn at Noon	Ebn at Noon	743	788	817	820	859	856	831	815	805	774	725	700		
		Edh at Noon	81	104	126	145	126	122	127	125	110	93	78	73		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.12	1.93	3.22	4.91	6.19	6.65	6.77	5.96	4.33	2.65	1.45	0.99		
		RadStd	0.13	0.24	0.28	0.48	0.52	0.45	0.34	0.34	0.33	0.26	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.73	N/A	N/A	+1.01	N/A	N/A	-118	-171	+156	+93

Nomenclature: See separate page